MANELISI NCWABA

BSc Applied Mathematics & Statistics Graduate
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SUMMARY

Motivated and detail-oriented BSc Applied Mathematics and Statistics graduate with hands-on Data Science internship experience, focusing on data analysis, data cleaning, visualization, and exploratory data analysis. Strong interest in data-driven decision-making and transforming complex datasets into meaningful insights using Python, SQL, and Excel. Aspiring to grow as a Data Analyst, Business Analyst, or Junior Data Scientist, contributing analytical thinking, statistical reasoning, and continuous learning to dynamic teams.

TECHNICAL SKILLS

Programming & Tools: Python, HTML, CSS, SAS, SQL

Data Science & Analysis: Pandas, NumPy, Matplotlib · Data Cleaning & Preprocessing · Exploratory Data Analysis (EDA) · Data Visualization · Basic Machine Learning (Regression, Classification, Decision Trees)

Business Intelligence: Power BI · Microsoft Excel (data analysis, formulas, reporting)

Version Control & Dev: GitHub · Visual Studio Code (VS Code)

Office & Collaboration: Microsoft Office 365 (Excel, Word, PowerPoint, Outlook, Teams)

PROJECTS

SME Financial Health Prediction | Zindi / data.org / FinMark Trust Challenge | *2026*

Machine learning pipeline to classify the financial health of SMEs across 4 Southern African nations (Eswatini, Lesotho, Zimbabwe, Malawi) as Low, Medium, or High — supporting improved credit access and policy decisions for 43M+ underserved businesses.

- Built a 6-stage end-to-end ML pipeline: data loading, feature engineering, country-grouped imputation, label encoding, SMOTE oversampling, and ensemble prediction
- Engineered 30+ domain-driven features including financial_access_score, insurance_score, profit_proxy, and expense_ratio to capture financial resilience beyond raw survey data
- Designed a weighted ensemble model (Random Forest 35% + Gradient Boosting 40% + Extra Trees 25%) achieving an 89% macro F1 score on the public leaderboard
- Applied SMOTE inside stratified k-folds to handle severe class imbalance (Low: 55%, Medium: 33%, High: 12%) without data leakage
- Per-class F1 scores: Low 0.91 · Medium 0.87 · High 0.88 — strong performance across all risk tiers
- Final model prediction achieved 89% F1 score using XGBoost, confirming the ensemble approach and validating the feature engineering pipeline

Tech stack: Python · Scikit-learn · Pandas · NumPy · imbalanced-learn (SMOTE) · GitHub

WORK EXPERIENCE

Data Science Intern | **Shaper** | Johannesburg, South Africa *Oct 2025 – Present*

- Performed data cleaning, analysis, and visualization using Python and Excel
- Conducted exploratory data analysis to identify trends and insights
- Applied statistical and analytical techniques to real-world datasets
- Created visual reports and dashboards to communicate findings
- Documented analysis and projects using GitHub

Waitress | **Spur** | Durban, South Africa *Jun 2024 – Nov 2024*

- Delivered exceptional customer service, managed billing (cash & card), and upsold menu items to boost sales
- Coordinated with kitchen staff to ensure timely, accurate order fulfilment

- Collaborated with team members during peak hours to maintain smooth operations

Academic Assistant | **Mehlokubheka High School** | Mtubatuba, KZN *Jan 2023 – Mar 2023*

- Assisted educators in delivering lessons with a focus on mathematics and science
- Provided one-on-one and group academic support; marked assessments and maintained grade records
- Supported learners with exam preparation, study techniques, and homework reinforcement

EDUCATION

BSc in Applied Mathematics and Statistics | University of KwaZulu-Natal | *2022 – 2024*

Design Thinking Practitioner (NQF Level 4) Learnership | *SpecCon Holdings* | *In Progress*

ACHIEVEMENTS

Full Stack Development – IT Varsity (FNB Academy), 2025